

DRY-RUN GUIDE | magic.py
Source: Python\3rd_Belt\Matrix\magic.py

CORE BEHAVIOR

Checks that every row, column, and both diagonals have the same sum.

REPRESENTATIVE INPUT

Lo Shu square: '8 1 6' / '3 5 7' / '4 9 2'.

STEP TRACE

Target row sum=15; every row, column, and diagonal sums to 15.

EXPECTED OUTPUT / RETURN

'True'

ACTUAL SOURCE EXCERPT

Verbatim excerpt from Python\3rd_Belt\Matrix\magic.py; line numbers refer to the original file. No source was modified.

```
001 | # Question: Check whether a square matrix is a magic square.
002 | # Add your solution here.
003 |
004 |
005 | def is_magic_square(matrix, N=None):
006 |     if not matrix or not matrix[0]:
007 |         return False
008 |
009 |     n = len(matrix) if N is None else N
010 |     if n <= 0:
011 |         return False
012 |
013 |     target_sum = sum(matrix[0])
014 |
015 |     for row in matrix:
016 |         if len(row) != n or sum(row) != target_sum:
017 |             return False
018 |
019 |     for col in range(n):
020 |         col_sum = sum(matrix[row][col] for row in range(n))
021 |         if col_sum != target_sum:
022 |             return False
023 |
024 |     diag1_sum = sum(matrix[i][i] for i in range(n))
025 |     if diag1_sum != target_sum:
026 |         return False
027 |
028 |     diag2_sum = sum(matrix[i][n - 1 - i] for i in range(n))
029 |     if diag2_sum != target_sum:
030 |         return False
031 |
032 |     return True
033 |
034 |
035 | if __name__ == "__main__":
036 |     n = int(input())
037 |     matrix = [list(map(int, input().split())) for _ in range(n)]
```

```
038 |     print(is_magic_square(matrix))
039 |
```

DRY-RUN NOTE

Values are representative. For function-only modules the guide uses an illustrative direct call and explains its return; quirks/errors describe actual behavior.